

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

A ball of mass 0.5 kg is attached to a string of length 50 cm. The ball is rotated on a horizontal circular path about its vertical axis. The maximum tension that the string can bear is 400 N. The maximum possible value of angular velocity of the ball in rad/s is,:

- (1) 1600
- (2) 40
- (3) 1000
- (4) 20

Q2 - 2024 (01 Feb Shift 2)

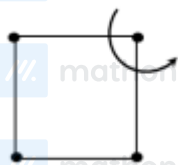
A disc of radius R and mass M is rolling horizontally without slipping with speed v . It then moves up an inclined smooth surface as shown in figure. The maximum height that the disc can go up the incline is :



- (1) $\frac{v^2}{g}$
- (2) $\frac{3}{4} \frac{v^2}{g}$
- (3) $\frac{1}{2} \frac{v^2}{g}$
- (4) $\frac{2}{3} \frac{v^2}{g}$

Q3 - 2024 (27 Jan Shift 1)

Four particles each of mass 1 kg are placed at four corners of a square of side 2 m. Moment of inertia of system about an axis perpendicular to its plane and passing through one of its vertex is _____ kgm^2 .



Q4 - 2024 (27 Jan Shift 2)

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Questions with Answer Keys

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A ring and a solid sphere roll down the same inclined plane without slipping. They start from rest. The radii of both bodies are identical and the ratio of their kinetic energies is $\frac{7}{x}$ where x is _____

Q5 - 2024 (29 Jan Shift 1)

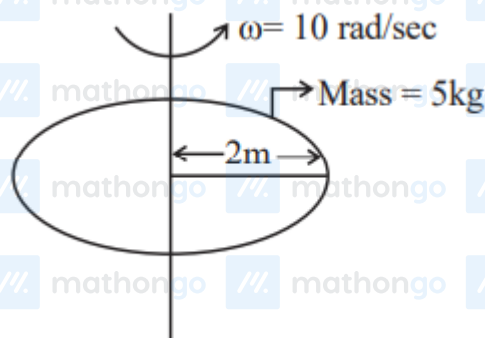
A cylinder is rolling down on an inclined plane of inclination 60° . Its acceleration during rolling down will be $\frac{x}{\sqrt{3}} \text{ m/s}^2$, where $x =$ _____ (use $g = 10 \text{ m/s}^2$).

Q6 - 2024 (29 Jan Shift 2)

A body of mass 5 kg moving with a uniform speed $3\sqrt{2} \text{ ms}^{-1}$ in X – Y plane along the line $y = x + 4$. The angular momentum of the particle about the origin will be _____ $\text{kgm}^2 \text{ s}^{-1}$.

Q7 - 2024 (30 Jan Shift 1)

Consider a Disc of mass 5 kg, radius 2 m, rotating with angular velocity of 10 rad/s about an axis perpendicular to the plane of rotation. An identical disc is kept gently over the rotating disc along the same axis. The energy dissipated so that both the discs continue to rotate together without slipping is _____ J.



Q8 - 2024 (30 Jan Shift 2)

Two discs of moment of inertia $I_1 = 4 \text{ kg m}^2$ and $I_2 = 2 \text{ kg m}^2$ about their central axes & normal to their planes, rotating with angular speeds 10 rad/s & 4 rad/s respectively are brought into contact face to face with their axis of rotation coincident. The loss in kinetic energy of the system in the process is _____ J.

Q9 - 2024 (31 Jan Shift 2)

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A body of mass ' m ' is projected with a speed ' u ' making an angle of 45° with the ground. The angular momentum of the body about the point of projection, at the highest point is expressed as $\frac{\sqrt{2}mu^3}{Xg}$. The value of 'X' is _____

Q10 - 2024 (31 Jan Shift 2)

Two identical spheres each of mass 2 kg and radius 50 cm are fixed at the ends of a light rod so that the separation between the centers is 150 cm. Then, moment of inertia of the system about an axis perpendicular to the rod and passing through its middle point is $\frac{x}{20} \text{ kg m}^2$, where the value of x is _____

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Answer Key

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Solutions

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Q1

$$T = m\omega^2 \ell$$

$$400 = 0.5\omega^2 \times 0.5$$

$$\omega = 40 \text{ rad/s.}$$

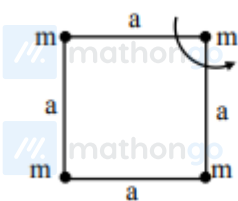
Q2

Only the translational kinetic energy of disc changes into gravitational potential energy. And rotational KE remains unchanged as there is no friction.

$$\frac{1}{2}mv^2 = mgh$$

$$h = \frac{v^2}{2g}$$

Q3



$$I = ma^2 + ma^2 + m(\sqrt{2}a)^2$$

$$= 4ma^2$$

$$= 4 \times 1 \times (2)^2 = 16$$

Q4

In pure rolling work done by friction is zero. Hence potential energy is converted into kinetic energy. Since initially the ring and the sphere have same potential energy, finally they will have same kinetic energy too.

$\therefore$ Ratio of kinetic energies = 1

$$\Rightarrow \frac{7}{x} = 1 \Rightarrow x = 7$$

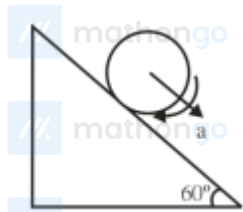
Q5

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Solutions

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For rolling motion, $a = \frac{g \sin \theta}{1 + \frac{I_{cm}}{MR^2}}$

$$a = \frac{g \sin \theta}{1 + \frac{1}{2}}$$

$$= \frac{2 \times 10 \times \frac{\sqrt{3}}{2}}{3}$$

$$= \frac{10}{\sqrt{3}}$$

Therefore $x = 10$

Q6

$$y - x - 4 = 0$$

d_1 is perpendicular distance of given line from origin.

$$d_1 = \left| \frac{-4}{\sqrt{1^2 + 1^2}} \right| \Rightarrow 2\sqrt{2} \text{ m}$$

So,

$$|\vec{L}| = mvd_1 = 5 \times 3\sqrt{2} \times 2\sqrt{2} \text{ kg m}^2/\text{s} \\ = 60 \text{ kg m}^2/\text{s}$$

Q7

$$\vec{L}_i = I\omega_i = \frac{MR^2}{2} \cdot \omega = 100 \text{ kg m}^2/\text{s}$$

$$E_i = \frac{1}{2} \cdot \frac{MR^2}{2} \cdot \omega^2 = 500 \text{ J}$$

$$\vec{L}_i = \vec{L}_f \Rightarrow 100 = 2I\omega_f$$

$$\omega_f = 5 \text{ rad/sec}$$

$$E_f = 2 \times \frac{1}{2} \cdot \frac{5(2)^2}{2} \cdot (5)^2 = 250 \text{ J}$$

$$\Delta E = 250 \text{ J}$$

Q8

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Solutions

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$$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega_0 \text{ (C.O.A.M.)}$$

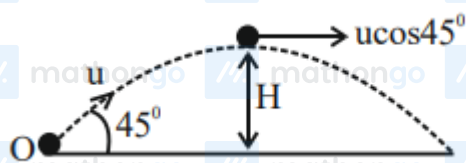
$$\text{gives } \omega_0 = 8 \text{ rad/s}$$

$$E_1 = \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 = 216 \text{ J}$$

$$E_2 = \frac{1}{2}(I_1 + I_2)\omega_0^2 = 192 \text{ J}$$

$$\therefore \Delta E = 24 \text{ J}$$

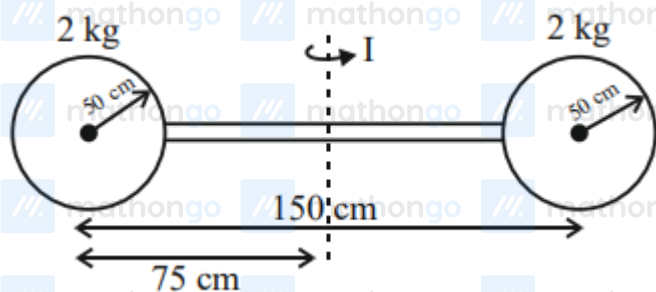
Q9



$$L = mu \cos \theta \frac{u^2 \sin^2 \theta}{2g}$$

$$= mu^3 \frac{1}{4\sqrt{2}g} \Rightarrow x = 8$$

Q10



$$I = \left(\frac{2}{5}mR^2 + md^2 \right) \times 2$$

$$I = 2 \left(\frac{2}{5} \times 2 \times \left(\frac{1}{2} \right)^2 + 2 \times \left(\frac{3}{4} \right)^2 \right) = \frac{53}{20} \text{ kg-m}^2$$

$$X = 53$$

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